Exam 2 In-Class Review

Module 6 - Multiple Dosings · Review set · 7 Oct 2026

Name _____ Date _____

Question 1 Using the patient's ideal body weight, estimate the creatinine clearance of a 45-year-old female who weighs 154 lb, serum creatinine of 0.78 mg/dL, and stands 5'8" tall.

Question 2 A drug was administered as a single IV bolus dose of 10 mg/kg to a 75-kg male volunteer. The drug has an elimination half-life of approximately 5 hours and volume of distribution of 2.3 L/kg. Urine samples were collected for 48 hours and upon analysis, 630 mg was recovered. What is the renal clearance in mL/min and probable mechanism of renal elimination?

Question 2 A drug was administered as a single IV bolus dose of 10 mg/kg to a 75-kg male volunteer. The drug has an elimination half-life of approximately 5 hours and volume of distribution of 2.3 L/kg. Urine samples were collected for 48 hours and upon analysis, 630 mg was recovered. What is the probable mechanism of renal elimination?

A. Filtration with tubular reabsorption
B. Filtration with active secretion
C. Filtration only

3a An antibiotic is to be administered as 12 mg/kg IV bolus doses every 8 hours to a 75-kg patient. The antibiotic has a total body clearance of approximately 1.86 L/hr and an apparent volume of distribution that is 15% of body weight. a. What is the expected maximum steady-state concentration of this antibiotic in this patient?

3b An antibiotic is to be administered as 12 mg/kg IV bolus doses every 8 hours to a 75-kg patient. The antibiotic has a total body clearance of approximately 1.86 L/hr and an apparent volume of distribution that is 15% of body weight. b. What dose would yield an average steady-state concentration of approximately 80 mcg/mL?

3c An antibiotic is to be administered as 12 mg/kg IV bolus doses every 8 hours to a 75-kg patient. The antibiotic has a total body clearance of approximately 1.86 L/hr and an apparent volume of distribution that is 15% of body weight. c. If the AUC following a 500 mg oral dose of the antibiotic was 202 (mg/L)hr, what is the oral bioavailability of this antibiotic?

3d An antibiotic is to be administered as 12 mg/kg IV bolus doses every 8 hours to a 75-kg patient. The antibiotic has a total body clearance of approximately 1.86 L/hr and an apparent volume of distribution that is 15% of body weight. d. What oral dose would provide the same extent of absorption as the 12 mg/kg IV dose?

3e An antibiotic is to be administered as 12 mg/kg IV bolus doses every 8 hours to a 75-kg patient. The antibiotic has a total body clearance of approximately 1.86 L/hr and an apparent volume of distribution that is 15% of body weight. e. What C_{max} would you expect following the administration of the first oral dose in the amount you determined in part d if the absorption half-life is 75 minutes?

3f An antibiotic is to be administered as 12 mg/kg IV bolus doses every 8 hours to a 75-kg patient. The antibiotic has a total body clearance of approximately 1.86 L/hr and an apparent volume of distribution that is 15% of body weight. f. What dosing interval would be appropriate to achieve an average steady-state concentration of 80 mcg/mL using the oral dose determined in part d?

3g An antibiotic is to be administered as 12 mg/kg IV bolus doses every 8 hours to a 75-kg patient. The antibiotic has a total body clearance of approximately 1.86 L/hr and an apparent volume of distribution that is 15% of body weight. g. What is t_{max} at steady state if the dose determined in part d were administered per the dosing interval determined in part f?

3h An antibiotic is to be administered as 12 mg/kg IV bolus doses every 8 hours to a 75-kg patient. The antibiotic has a total body clearance of approximately 1.86 L/hr and an apparent volume of distribution that is 15% of body weight. h. What is C_{max} if the dose determined in part d were administered per the dosing interval determined in part f?

Question 4 Four hundred milligrams of an antibiotic was infused intravenously over a period of 2 hours. Eight hours after the start of the first infusion, a second 400-mg dose was infused again over a period of 2 hours. The half-life of the antibiotic is approximately 3.75 hours and the apparent volume of distribution is approximately 22 L. What is the plasma drug concentration 4 hours after the cessation of the second infusion?

Question 5 What is creatinine clearance used to estimate?

A. renal function
B. metabolic function
C. tubular reabsorption
D. apparent volume of distribution

Question 6 Which of the following would you expect with a 2-fold increase in oral dose? (Select all that apply)

A. Decreased AUC
B. Increased AUC
C. Decreased t_{max}
D. Decreased C_{max}
E. Increased t_{max}
F. Increased C_{max}

Question 7 A new antibiotic is excreted by the kidney and has a renal clearance that is approximately 300 mL/min. What is the probable mechanism of renal excretion of this drug?

A. Filtration with tubular reabsorption
B. Filtration with active secretion
C. Filtration only

Question 8 Which term describes the rate and extent to which the active ingredient or active moiety is absorbed from a drug product and becomes available at the site of action?

A. pharmacodynamics
B. bioavailability
C. product performance
D. bioequivalence

Question 9 Which of the following would be expected following a decrease in the dosing interval with no change to the dose?

A. increased steady-state concentrations
B. increased patient compliance
C. increased fluctuations between peak to trough concentrations
D. decreased MTC

Question 10

The principle of superposition assumes that early doses of drug have no effect on the pharmacokinetics of subsequent doses.

A. True
B. False

Question 11

Which term below describes a comparison of the bioavailability of the active drug in the systemic circulation following extravascular administration with the bioavailability of the same drug following intravenous administration?

A. drug product performance
B. bioequivalence
C. relative bioavailability
D. absolute bioavailability

Question 12

Which of the following is true regarding the absorption phase following oral administration?

A. drug at the absorption site has become depleted
B. the rate of drug absorption is faster than the rate of drug elimination
C. the rate of drug elimination is faster than the rate of drug absorption
D. the rate of drug absorption equals the rate of drug elimination

Question 13

When does the minimum concentration of drug in the plasma occur with multiple IV bolus dosings?

A. at the end of the dosing interval
B. in 3 to 5 half-lives
C. when the rate of absorption exceeds the rate of elimination
D. when the rate of absorption equals the rate of elimination

Question 14

If there is an decrease in the total body clearance of an agent due to renal dysfunction, what is the expected change to the elimination half-life of that agent?

A. the elimination half-life will increase
B. the elimination half-life will decrease
C. the elimination half-life will not change